

18.11.2024 (M)
Comp



SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2024 –November 2024			
Maximum Marks: 100		Examination: ESE Examination	
		Duration:3 Hrs.	
Programme code:01		Class: _LY__	Semester: VII_(SVU 2020)
Programme: B. Tech. (Computer Engineering)			
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: COMP	
Course Code: 116U01C701		Name of the Course: Software Architecture & Design Thinking	
Instructions:			
1) Draw neat diagrams			
2) All questions are compulsory			
3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any FOUR	20
i)	Explain Object Oriented Software Architecture Style with suitable example.	5
ii)	What is viewpoint? List the different type of the viewpoints in Software Architecture and explain.	5
iii)	Explain _____ Components with Examples	5
iv)	What are Analysis Goals?	5
v)	Explain different types of inconsistencies in the views.	5
vi)	What is a Software Connector? What are the reasons for Treating Connectors Independently?	5
Q2 A	Solve the following:	10
i)	Explain Model View Controller Architecture with example.	5
ii)	What do you understand by Architecture Style. Give the benefits of using the Architecture Style.	5
	OR	
Q2 A	Explain Data Flow style and give the architecture of Banking System in Data Flow style.	10
Q 2 B	Solve any ONE	10
i)	Compare and contrast Data Access Connector and Stream Connector in detail.	10
ii)	Explain Architectural Modelling. List out the various component that can be modelled in detail.	10
Q3	Solve any TWO	20
i)	Explain different steps in the research stage of design thinking.	10
ii)	What are the key forms of appropriation of refinement stage in design thinking?	10
iii)	With reference to “thinking” stage of design thinking, explain the various themes of thinking.	10

Que. No.	Question	Max. Marks
Q4	Solve any TWO	20
i)	Explain _____ the different Vocabulary styles used in “Prototyping” stage of	10

	design thinking with the help of examples.	
ii)	Explain different types of the Prototype used in the Thinking Process. Justify their usage in various types of system using example?	10
iii)	Apply the various stages of Design Thinking in Railway Reservation System like IRCTC.	10
Q5	Write short notes on any FOUR	20
i)	Various ways of appropriation in designing a system	5
ii)	Thinking in signs in Refinement	5
iii)	Semiotics, denotation and connotation in thinking	5
iv)	Prescriptive v/s descriptive Software architecture	5
v)	ATAM Analysis	5
vi)	Design decisions in the development of software	5